WiFi-TTA-Bench: Test-Time Adaptation Under Physics-Structured Wireless Shift

Anonymous Author(s)
Affiliation
Address
email

Abstract

We present **WiFi-TTA-Bench**, a harm-aware benchmark for test-time adaptation (TTA) on WiFi Channel State Information (CSI) under *physics-structured* domain shift. Our headline finding is a clean counterexample against the "TTA always helps" and the "TTA always fails" narratives: the *same* BatchNorm-equipped MLP+BN backbone with faithful BN-only TENT achieves $+21.1$ pp on SignFi's milder temporal split but -0.61 pp on Widar_BVP's harder cross-room shift (n=15 paired observations each). This suggests that on the evaluated WiFi CSI splits, TTA outcomes depend on the interaction between shift severity and architecture rather than on the choice of adaptation method alone. On the primary MLP backbone, no evaluated method's 95% bootstrap CI clears source-only inference on any of the three real datasets, and our simplified shared-MLP ports of SAR and CoTTA are strongly negative across all three datasets (Widar -7.6/-6.6, NTU-Fi -17.7/-17.7, SignFi -11.8/-12.4 pp), while faithful BN-only SAR/CoTTA ports on an MLP+BN Widar backbone shrink to -0.15/-0.38 pp with CIs overlapping zero. The benchmark covers up to 12 methods on Widar_BVP (10–12 on the other datasets; coverage matrix in Appendix B), with explicit fidelity labels (TENT-proj vs faithful BN-only TENT; SHOT as IM; simplified SAR/CoTTA/LAME ports). We present evidence consistent with three failure mechanisms (overconfident target predictions with mean max-softmax confidence 70–81 % across every class despite per-class accuracies as low as 24 %; batch-statistic unreliability at WiFi-scale batches; and a large synthetic-to-real swing for the physics-prior objective family), and show that source calibration via label smoothing reduces `physics_tta` damage on Widar (not significant at n=15). WiFi-TTA-Bench delivers standardized splits, machine-readable metadata (Croissant core + RAI fields), a harm-aware 5-metric evaluation protocol (gain, CI, Cohen's d, negative-adaptation rate, source drop), and a tested artifact (248 automated tests) with component scripts that regenerate every main table and figure from prepared data.

1 Introduction

WiFi Channel State Information (CSI) sensing—activity recognition, gesture detection, localization using wireless signals—is a growing field with a well-documented deployment problem: models trained in one room fail in another, with accuracy drops of 30–50% commonly reported Yousefi et al. [2017]. Test-time adaptation (TTA) is the natural solution: WiFi infrastructure provides continuous unlabeled data from the target environment, exactly matching TTA's problem formulation.

But does TTA actually work for WiFi CSI? This question has not been systematically studied. Prior WiFi sensing work focuses on supervised domain adaptation or domain generalization, skipping the label-free deployment-time regime entirely. Meanwhile, TTA research evaluates almost exclusively

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

on vision corruption benchmarks (ImageNet-C, CIFAR-C), where shifts are statistical rather than physical.

We fill this gap with **WiFi-TTA-Bench**, a WiFi CSI benchmark designed to answer whether TTA methods that succeed on vision benchmarks also succeed on *physics-structured* domain shifts—shifts caused by changes in the physical measurement process (multipath propagation) rather than surface-level corruptions.

Headline counterexample. On Widar's hard cross-room shift, faithful BN-only TENT on an MLP+BN backbone averages -0.61 pp (n=15, Table 6). On SignFi's milder temporal split, the *same* BN-only TENT on the *same* MLP+BN backbone averages $+21.1$ pp (n=15, 95%CI $= [+13.1, +29.6]$, Table 7). On Widar, no method evaluated on the primary MLP backbone has a 95% CI clearing zero, and our simplified shared-MLP ports of SAR/CoTTA show strongly negative mean gain on all three real datasets; their faithful BN-only counterparts shrink toward zero. This pattern suggests that, on the evaluated WiFi CSI splits, the locus of TTA outcome depends on the interaction between shift severity, architecture, and source calibration rather than on the choice of adaptation method alone.

Evaluative role. Following the NeurIPS 2026 Evaluations & Datasets track framing, WiFi-TTA-Bench is first an evaluation artifact rather than a method contribution. We state its evaluative role explicitly: the benchmark tests the hypothesis that existing TTA methods—evaluated to date almost exclusively on vision corruption benchmarks—transfer to WiFi CSI under environmental and temporal shifts. It supports claims of the form "under leave-one-environment-out splits on {Widar_BVP, NTU-Fi, SignFi-10} with the shared MLP or MLP+BN backbones, method M achieves mean gain g with CI $[l, u]$ and negative-adaptation rate r." Its scope does not extend to cross-modality generalisation, device-in-the-loop latency, or real-time deployment. Its primary limitation is that it does not collect new raw CSI and therefore inherits the three upstream datasets' scale and split characteristics (two of three are temporal splits within a single laboratory).

Contributions.

1. **Empirical finding: evidence suggests a shift-severity $\times$ architecture interaction on WiFi CSI.** The SignFi/Widar counterexample above rules out a universal "TTA fails on physics-structured shift" interpretation while still supporting the weaker claim that no evaluated method achieves a positive CI on the primary MLP backbone on any real dataset.

2. **Harm-aware benchmark** with 12 methods on Widar (10–12 on NTU-Fi/SignFi; coverage matrix in Appendix B), 3 real datasets, synthetic control, 15 paired observations per method, explicit fidelity labels (TENT-proj vs faithful BN-only TENT; SHOT as IM; simplified SAR/CoTTA/LAME), and a 5-metric protocol (gain, CI, Cohen's d, negative-adaptation rate, source drop).

3. **Diagnostic evidence consistent with three failure modes**: (i) overconfident target predictions—source-model mean max-softmax confidence is 70–81 % across every Widar class while per-class accuracies range from 24 % to 57 % (Appendix Table 11); (ii) batch-statistic unreliability at WiFi-scale batches (faithful BN-only TENT -0.61 pp, BN-reset -1.11 pp on Widar); (iii) a large synthetic-to-real swing for the physics-prior objective family ($+7.4$ to $+8.0$ on synthetic $\rightarrow$ -0.5 pp on real Widar). Label smoothing (α=0.1) on Widar gives a small, non-significant harm reduction (negative-adaptation rate 0.47 $\rightarrow$ 0.27 for the best gradient-based method; Appendix H); a labelled-target oracle upper bound ($+24.9$ pp mean gap over 3 rooms $\times$ 3 seeds, `scripts/train_supervised_upper_bound.py`; Appendix P) quantifies that unlabelled TTA under this protocol recovers $\approx$1% of the available headroom.

4. **Reproducible artifact**: code at `https://anonymous.4open.science/r/wifi-tta-bench-ed2026`, mirrored Croissant 1.0 dataset card at `https://huggingface.co/datasets/anonymous-ed2026/wifi-tta-bench`, data-preparation scripts, split manifests, a fail-fast reproducer (`scripts/reproduce_all_results.py`), and 248 automated tests (both URLs anonymised for double-blind review).

2 Related Work

TTA benchmarks and methods. TTA evaluation is dominated by vision corruption benchmarks: ImageNet-C Hendrycks and Dietterich [2019], CIFAR-C, WILDS Koh et al. [2021], BREEDS Santurkar et al. [2021], and Shifts Malinin et al. [2021]. Representative methods include TENT Wang et al. [2021a], SHOT Liang et al. [2020], T3A Iwasawa and Matsuo [2021], CoTTA Wang et al. [2022b], SAR Niu et al. [2023], and parameter-free LAME Boudiaf et al. [2022]; an expanded method inventory (TTT/TTT++/NOTE/RoTTA/EATA/AdaContrast) is deferred to Appendix A. Recent work questions TTA's generality: Zhao et al. Zhao et al. [2023] catalog pitfalls, Press et al. Press et al. [2023] show a "dumb" baseline competes with sophisticated methods, and Su et al. Su et al. [2024] address real-world instability. All assume covariate shift Shimodaira [2000], Quiñonero-Candela et al. [2009], which we argue WiFi room-to-room shift violates.

WiFi CSI and domain adaptation. Standard benchmarks—SignFi Ma et al. [2018], UT-HAR Yousefi et al. [2017], Widar3.0 Zheng et al. [2019], SenseFi Yang et al. [2023a]—and supervised or source-target domain-adaptation methods—EI Jiang et al. [2018], RF-Net Ding et al. [2020], AirFi Wang et al. [2022a], AutoFi Yang et al. [2023b], DANN Ganin et al. [2016], CDAN Long et al. [2018]—address cross-environment robustness, but none evaluates label-free deployment-time adaptation. Calibration remains a known source of overconfidence Guo et al. [2017], Müller et al. [2019].

Physics-informed wireless. PINNs Raissi et al. [2019], Karniadakis et al. [2021] embed PDEs as losses and can exhibit gradient pathologies Wang et al. [2021b], Krishnapriyan et al. [2021]. Recent work applies physics priors to channel estimation Javid and González Prelcic [2025] and radio mapping Böck et al. [2025]. We evaluate whether an OFDM-derived residual prior can serve as a TTA signal.

3 Physics-Guided TTA Variants

This section defines the three physics-guided variants evaluated in the benchmark. They are not the paper's main contribution; they exist to test whether a coarse OFDM residual prior can make TTA less harmful than generic entropy minimisation. The objective is *feature-statistic alignment using a coarse OFDM residual-prior representation*, not a direct PDE residual loss and not a channel-estimation method.

Residual-prior representation. For an input CSI observation $\mathbf{x}$, we compute an analytical prior $\mathbf{p}(\mathbf{x})$ from the synthetic OFDM propagation model and feed the encoder the concatenation $\tilde{\mathbf{x}} = [\mathbf{x}, \mathbf{p}(\mathbf{x}), \mathbf{x} - \mathbf{p}(\mathbf{x})]$. The prior is intentionally coarse on real data because the true path gains, delays, and room geometry are unobserved. The classifier head is frozen during adaptation; only the encoder-side parameters listed in Appendix M are updated.

Feature-statistic alignment objective. After source training, we store source feature mean/variance $(\boldsymbol{\mu}_s, \boldsymbol{\sigma}_s^2)$ on the encoder representation $\mathbf{z} = f_\phi(\tilde{\mathbf{x}})$. In the implementation the same $\mathbf{z}$ is used for both the primary and auxiliary moment terms (`outputs["features"]`); we reserve the word "residual-branch" for the input concatenation and do not claim a separate internal residual stream. We also store auxiliary statistics $(\boldsymbol{\mu}_s^r, (\boldsymbol{\sigma}_s^r)^2)$. For an unlabelled target batch B_t the objective is

$$\mathcal{L}_{\text{align}} = \|\boldsymbol{\mu}_t - \boldsymbol{\mu}_s\|_2^2 + \|\boldsymbol{\sigma}_t^2 - \boldsymbol{\sigma}_s^2\|_2^2 + \lambda_r \big(\|\boldsymbol{\mu}_t^r - \boldsymbol{\mu}_s^r\|_2^2 + \|(\boldsymbol{\sigma}_t^r)^2 - (\boldsymbol{\sigma}_s^r)^2\|_2^2\big)\,,$$

with $\lambda_r = 1$. `physics_tta` applies this objective under the standard budget. `safe_physics` rejects a step when the unlabelled objective increases or feature-statistic drift exceeds $\tau_a = 2.0$. `selective_physics` softly blends the adapted and source predictions, $q(y|\mathbf{x}) = g(\mathbf{x})q_{\text{adapt}} + (1-g(\mathbf{x}))q_{\text{src}}$, with $g(\mathbf{x}) = \text{clip}((c_{\text{adapt}} - c_{\text{src}})/\tau_c + 0.5, 0, 1)\exp(-d_{\text{align}}/\tau_a)$ where c is max softmax confidence and d_{align} is feature-statistic divergence from the source reference. Thresholds (τ_c=0.2, τ_a=2.0) were fixed before evaluation.

Table 1: Datasets and methods. The **Shape** column reports $(\text{features}, \text{time/packets})$ per sample for the prepared CSI tensors used by the benchmark; the SignFi-10 row reports $(\text{flattened features}, \text{prepared dim})$. Full dataset protocol, per-method hyperparameters, and absolute baselines are in Appendix M/J.

Dataset	Samples	Classes	Envs / Shape	Shift Type
Widar_BVP Zheng et al. [2019]	8,964	6 gestures	3 rooms / $(22, 400)$	Cross-room
NTU-Fi HAR	1,200	6 activities	3 sessions / $(114, 500)$	Cross-session (temporal)
SignFi-10 Ma et al. [2018]	200	10 gestures	3 splits / $(200, 90)$	Cross-split (temporal)
Synthetic	config.	3	2 domains / $(8, 2)$	Controlled physics

4 WiFi-TTA-Bench: Benchmark Design

Protocol. Leave-one-environment-out for each real dataset; 3 folds $\times$ 5 seeds $=$ 15 paired observations per method. Synthetic data uses the OFDM channel model $H(f_k) = \sum_l h_l \exp(-j2\pi f_k \tau_l)$ with class-specific multipath parameters and tunable shift severity (mild/moderate/strong). NTU-Fi sessions and SignFi splits are temporal splits within a single laboratory, not verified physical-room changes (see Limitations).

Methods (12 total: 11 TTA + `no_adapt`). Generic: `entropy_tta`, `cons. entropy`, `tent-proj` Wang et al. [2021a], `im` (SHOT variant) Liang et al. [2020], `t3a` Iwasawa and Matsuo [2021], `sar` Niu et al. [2023], `cotta` Wang et al. [2022b], `lame` Boudiaf et al. [2022]. Physics (ours): `physics_tta`, `safe_physics`, `selective_physics`. A faithful BN-affine `tent` is added on MLP+BN for Section 5.6.

Backbone. MLP encoder (3 layers $\times$ 64 hidden, latent dim 32) with residual-prior input $[\mathbf{x}, \mathbf{p}(\mathbf{x}), \mathbf{x} - \mathbf{p}(\mathbf{x})]$; classifier head frozen during adaptation. Architecture ablation with CNN1D+BN and MLP+BN in Section 5.6.

Implementation fidelity. The primary MLP has no BatchNorm, so TENT is implemented as projection-layer-only updates (`tent-proj`)—TENT's minimal-update principle without its BN-specific mechanism. SHOT is realised as entropy minimisation $+$ class-diversity regularisation (information maximisation, `im`) without pseudo-label self-training. SAR, CoTTA, and LAME are adapted to the shared backbone. Only the MLP+BN control in Section 5.6 is a faithful BN-affine TENT. Selective-physics thresholds (τ_c=0.2, τ_a=2.0) were fixed pre-evaluation.

Evaluation. Five metrics per method: mean target gain, 95% bootstrap CI (10K resamples), Cohen's d vs. `no_adapt`, negative adaptation rate $P(\text{gain} < 0)$, source drop. With 11 TTA methods vs. `no_adapt`, applying Bonferroni correction ($\alpha_{\text{corr}} \approx 0.0045$) does not change the qualitative conclusion, since no method's uncorrected 95% CI on a real dataset is positive on the primary backbone. Gain is reported as a fraction in detailed tables (0.003=0.3 pp) and in pp in prose. Source-only accuracy: Widar_BVP 37.3%, NTU-Fi 69.7%, SignFi-10 ≈95% (see Appendix J); the Widar cross-room gap is 52.1 pp.

5 Baseline Results

5.1 Main Finding: Primary-Backbone TTA Failure on Real WiFi CSI

Table 2 presents the central primary-backbone finding: *on the primary MLP backbone, no method's 95% CI clears zero on any of the three real datasets*. Entropy-based methods incur -13.8 to -15.1 pp mean source-domain degradation on Widar, while `selective_physics` keeps source drop to -1.5 pp. The cross-dataset rank correlation across the seven methods evaluated on all three datasets has mean Spearman $\rho \approx 0.70$ (computed by `scripts/analyze_tta_adaptation.py`). Per-method and per-room Widar tables, the leaderboard, step-budget and threshold-sensitivity studies, and the confusion-matrix visualisation (Figure 3, previously in main) are in Appendices D, E, F, and O.

Table 2: Decision table (primary MLP backbone, n=15 per method). For each dataset we report source-only accuracy, the best and worst TTA method (mean gain and 95% bootstrap CI for best; gain only for worst), the negative-adaptation rate of the best method, and the source-domain drop of that method. On the primary MLP, no method's 95% CI clears zero on any real dataset; `selective_physics` is consistently least harmful. Full per-method tables per dataset are in Appendix D.

Dataset	Backbone	no_adapt	Best TTA (gain, 95% CI)	Worst TTA (gain)	Neg rate	Src drop
Widar_BVP	MLP	37.3%	`selective_phys` +0.3 pp [−0.3, +0.8]	`sar` −7.6 pp	0.33	−1.5 pp
NTU-Fi HAR	MLP	69.7%	`tent-proj` −0.6 pp [−1.3, +0.1]	`entropy` −18.9 pp	0.53	−0.1 pp
SignFi-10	MLP	∼95%	`selective_phys` −0.8 pp [−2.0, +0.0]	`im` −12.4 pp	0.13	—
Synthetic	MLP	—	`safe_physics` +8.0 pp	`entropy` ≈0 pp	0.07	—

5.2 Cross-Dataset SAR / CoTTA / LAME and the Simplified-vs-Faithful Port

Recent TTA methods (SAR, CoTTA, LAME) rely on BatchNorm-affine updates or parameter-free output adjustment; porting them to our shared primary-MLP backbone required simplifications. To test whether the simplifications drive the negative results we add (i) the simplified-MLP port on all three datasets and (ii) the faithful BN-only SAR/CoTTA and full TENT on an MLP+BN backbone on Widar_BVP at n=15.

Table 3: Cross-dataset SAR / CoTTA / LAME (n=15). The faithful MLP+BN block shows that the large −7.6 / −6.6 pp simplified-MLP numbers on Widar partly reflect the port, not a universal method failure; the faithful numbers shrink to −0.15 / −0.38 pp and overlap zero, but still do not exceed the BN-only TENT control that also fails. LAME in our shared-backbone port is parameter-free and leaves predictions unchanged. Full table including NegRate in Appendix B.

Dataset / Port	Method	Gain (pp)	95% CI (pp)	NegRate
Widar_BVP (MLP, simplified)	sar	−7.62	[−9.86, −5.43]	0.93
	cotta	−6.58	[−9.42, −3.78]	0.80
	lame	0.00	[0.00, 0.00]	0.00
Widar_BVP (MLP+BN, faithful)	sar	−0.15	[−0.69, +0.43]	0.73
	cotta	−0.38	[−0.84, +0.12]	0.73
	tent (faithful, BN-only)	−0.01	[−0.37, +0.35]	0.47
NTU-Fi (MLP, simplified)	sar	−17.69	[−21.43, −14.18]	1.00
NTU-Fi (MLP, simplified)	cotta	−17.67	[−20.71, −14.46]	1.00
SignFi-10 (MLP, simplified)	sar	−11.81	[−17.90, −6.51]	0.73
SignFi-10 (MLP, simplified)	cotta	−12.38	[−18.32, −6.92]	0.73

The faithful-port rows answer the main baseline-fairness objection to the decision table: on an MLP+BN Widar backbone, SAR and CoTTA are neither harmful nor helpful (CIs overlap zero); faithful BN-only TENT on the same backbone is essentially zero (−0.01 pp). The SignFi MLP+BN counterexample (+21.1 pp; Table 7) therefore cannot be attributed to SAR/CoTTA being mis-ported, and the larger negative numbers on the simplified primary MLP port remain a legitimate harm-aware signal for deployers who cannot change the backbone.

5.3 Per-Class Overconfidence Evidence

Table 4 anchors the overconfidence claim in the intro and Mechanism 1: the source model assigns $\geq$ 70 % mean confidence to every Widar class while per-class accuracy varies from 24 % (`draw-zigzag`) to 57 % (`push&pull`), with the largest accuracy→confidence gap being 46 pp on `draw-zigzag`.

5.4 Source Calibration: Label Smoothing Ablation

At α=0.1, `physics_tta` damage is reduced (−0.20 → −0.10 pp) and negative adaptation rate drops from 0.60 to 0.47. The effect is modest and not statistically significant, but is directionally

Table 4: Per-class target accuracy and mean max-softmax confidence on Widar_BVP under the primary MLP backbone, aggregated across 15 source-only runs (3 rooms $\times$ 5 seeds). Confidence exceeds 70 % for every class while accuracy falls below 30 % for three of six classes. Artifact: `outputs/overconfidence/widar_per_class.json`.

Class	**Accuracy (%)**	**Mean confidence (%)**	**Conf−Acc gap (pp)**	n **(agg)**
slide	43.7	76.2	32.5	3669
draw-zigzag	23.8	70.4	**46.6**	3762
push&pull	57.4	80.9	23.5	3798
sweep	26.2	70.2	44.0	3642
clap	31.6	70.0	38.4	3816
draw-circle	28.5	71.4	42.9	3723

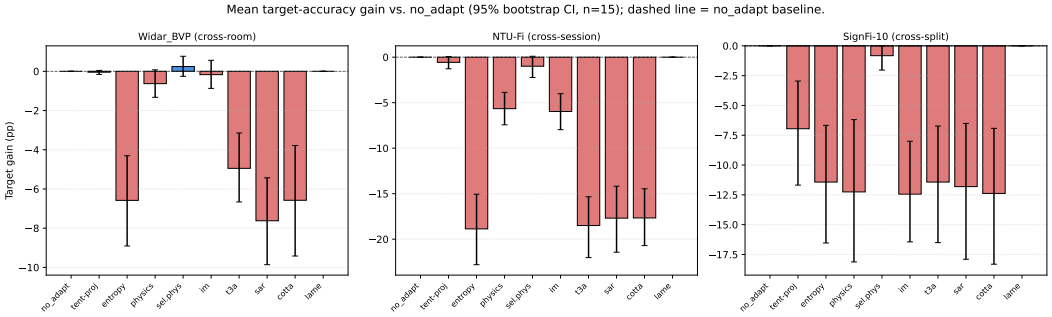

Figure 1: Mean target-accuracy gain (pp, 95% bootstrap CI) for all benchmark methods across Widar_BVP, NTU-Fi, and SignFi-10 on the primary MLP backbone (n=15 per cell). Dashed line: `no_adapt`. Bars below zero indicate negative adaptation. The *faithful* BN-only TENT counterexample on MLP+BN (Table 7) is not in this figure because the primary-MLP port of TENT is `tent-proj`; the counterexample is examined separately in Section 5.6.

consistent with calibration literature Guo et al. [2017], Müller et al. [2019]: reducing overconfidence limits the damage from entropy-based adaptation.

5.5 Supervised Upper Bound and Synthetic-to-Real Gap

A labelled-target oracle fine-tuned on Widar_BVP (per held-out room, 5-fold CV, Adam lr=1e-3, 30 epochs of fine-tuning from the matched source checkpoint; 3 rooms $\times$ 3 seeds $=$ 9 room/seed aggregates; `scripts/train_supervised_upper_bound.py`; Appendix P) improves mean target accuracy from 36.2% (matched source-only) to 61.1% under this protocol, a +24.9 pp supervised gap. The per-room gaps are strongly heterogeneous (+42.8 on room 2, +26.1 on room 0, +5.2 on room 1), which itself supports the shift-severity interpretation. The best unlabelled TTA gain on the primary MLP is +0.25 pp (`selective_physics`), so unlabelled TTA in this protocol recovers $\approx$1% of the supervised headroom. On controlled synthetic data with known physics parameters, `safe_physics` achieves +8.0 pp (d= + 0.72); the same physics-prior objective family yields −0.5 pp on real Widar (an $\approx$8.5 pp synthetic-to-real swing; Figure 2).

5.6 Architecture: Faithful TENT Controls and SignFi Counterexample

A key question is whether TTA's failure is due to the MLP's lack of BatchNorm. We implement `mlp_bn` (MLP with BatchNorm in every layer) and run faithful TENT (BN-affine-only updates) alongside BN-reset. Table 6 summarises the three-way Widar comparison; Table 7 gives the SignFi counterexample on the same MLP+BN backbone.

Key finding. The *same* BN-equipped backbone fails on Widar (−0.61 pp) and succeeds on SignFi (+21.1 pp). BN availability alone does not explain the hard-shift failure: TTA outcomes depend on the interaction between shift severity (Widar 52 pp domain gap vs. SignFi 14 pp), source calibration, and architecture. Per-row MLP vs MLP+BN means/stds (the full `arch_3way` table) and the one-room CNN1D+BN probe are reported in Appendix G/G.

Table 5: Label smoothing ablation on Widar_BVP (n=15). Smoothing α=0.1 reduces `physics_tta` damage from -0.20 to -0.10 pp and lowers the negative adaptation rate from 0.60 to 0.47.

Smoothing α	**Baseline**	**physics gain**	**tent gain**	**Neg Rate (phys)**
0.0 (none)	35.5%	-0.20 pp	$+0.01$ pp	0.60
0.1	35.9%	-0.10 pp	$+0.00$ pp	0.47
0.2	35.9%	-0.36 pp	-0.04 pp	0.60

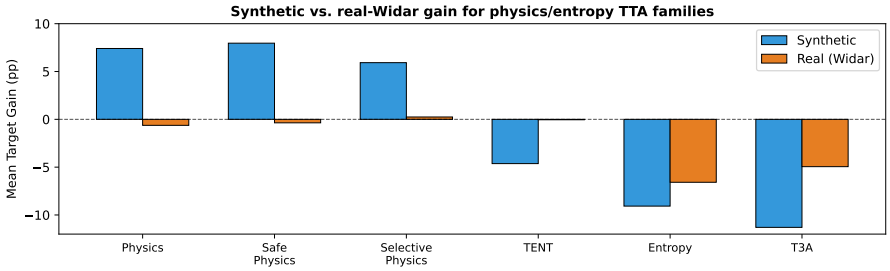

Figure 2: Synthetic vs. real gain for key methods. Physics-guided methods succeed with known parameters (blue) but fail on real data (orange).

6 What TTA Failure Reveals About Physics-Structured Shifts

6.1 Defining Physics-Structured Shift

We formalize the shift class that WiFi-TTA-Bench targets. Let $\mathbf{X} = G(\theta, \epsilon)$ be the data-generating process, where $\theta \in \Theta$ are physical parameters (path gains h_l, delays τ_l, room geometry) and ϵ is observation noise. A **physics-structured shift** occurs when the environment changes $\theta \to \theta'$ while the governing equation G remains invariant:

$$\text{Source: } \mathbf{X}_s = G(\theta_s, \epsilon), \quad \text{Target: } \mathbf{X}_t = G(\theta_t, \epsilon), \quad G \text{ unchanged.} \tag{1}$$

This differs from the standard covariate-shift assumption ($P(\mathbf{X})$ changes, $P(Y|\mathbf{X})$ preserved) and from corruption shift (additive perturbation to $\mathbf{X}$). In physics-structured shift, the mechanism generating features changes through unmeasured physical parameters, which can alter class-conditional structure. Our operational test for this paper is a cross-environment split in which room/session parameters change while the sensing task and label set are held fixed.

Table 8 quantifies this distinction:

The paradox—high variance but lower accuracy drop—is consistent with physics-structured shift: features scatter widely in patterns governed by room geometry rather than by independent random corruptions.

6.2 Why Entropy Minimization Can Fail Under Physics-Structured Shift

Standard TTA is usually motivated by covariate shift: $P_t(\mathbf{X}) \neq P_s(\mathbf{X})$ but $P_t(Y|\mathbf{X}) = P_s(Y|\mathbf{X})$. Under physics-structured shift, the data-generating process $G(\theta, \epsilon)$ changes through θ, which can alter both $P(\mathbf{X})$ and the effective class-conditional distributions. In that regime entropy minimisation reinforces the model's source-trained decision surface on target inputs even when that surface is misaligned; combined with overconfident source predictions (86–94% confidence at 10% accuracy), each entropy step can move predictions further from the target truth.

6.3 Three Mechanisms of TTA Failure

Evidence is consistent with three mechanisms, each with quantitative support. Hard physical-shift interpretations apply specifically to Widar_BVP, whose environmental splits are verified rooms; NTU-Fi and SignFi use temporal/session splits and are treated as softer evidence.

Table 6: Architecture ablation on real Widar_BVP. MLP and MLP+BN rows use n=15 paired observations (3 folds $\times$ 5 seeds); the CNN1D+BN row is an exploratory one-room probe with 3 seeds and is reported only as evidence that BN architectures do not automatically rescue TTA. MLP+BN raises the source-only baseline but no evaluated TTA method improves upon it within measurement noise; faithful BN-only TENT averages -0.61 pp on Widar.

Backbone	**Method**	**Gain (pp)**	**Abs Acc**
MLP (no BN)	`no_adapt`	0.00	35.7%
	`tent-proj`	$+0.01$	35.7%
	`physics`	-0.20	35.5%
MLP+BN	`no_adapt`	0.00	**36.9%**
	`bn_reset`	-1.11	35.8%
	`tent` (faithful, BN-only)	-0.61	36.4%
	`physics`	-0.55	36.4%
CNN1D+BN (real, 1 room)	`tent-proj`	$+2.63$	—
	`physics`	$+1.61$	—

Table 7: MLP+BN on SignFi-10 (n=15): the same backbone that fails on Widar's hard cross-room shift succeeds dramatically on SignFi's milder shift.

Method	**Gain (pp)**	**95% CI**	**Neg Rate**
`tent` (faithful, BN-only)	**+21.1**	$[+13.1, +29.6]$	**0.00**
`entropy`	$+15.6$	$[+8.2, +22.8]$	0.07
`physics`	$+12.3$	$[+4.8, +20.2]$	0.13

Mechanism 1: Overconfident misclassification. On Widar target rooms, aggregated across 15 runs, the source model's mean max-softmax confidence is 70–81 % across every class while per-class accuracy ranges from 24 % (draw-zigzag) to 57 % (push&pull) (Appendix Table 11). Entropy-minimising adaptation reinforces these over-confident and often incorrect predictions, consistent with the 0.93–1.00 negative-adaptation rates we observe for entropy-family methods on Widar. The step-budget analysis (Appendix F) shows monotonic degradation, consistent with each step pushing predictions further into incorrect modes.

Mechanism 2: Batch-statistic unreliability. Faithful TENT on MLP+BN achieves -0.61 pp (Table 6), which is consistent with BatchNorm plus BN-affine-only updates being insufficient on Widar_BVP. BN-reset (forward pass only) is actively harmful (-1.11 pp), which is consistent with target running statistics diverging from source statistics at WiFi scale (400–3,000 samples per environment, 32–64 per batch).

Mechanism 3: Physics-parameter dependence. The large swing (synthetic $+7.4$ to $+8.0$ $\rightarrow$ real -0.5 pp for the `physics_tta` objective family) is consistent with physics-guided adaptation working when parameters (h_l, τ_l) are observable but failing without them, under our protocol. This motivates channel-estimation-integrated TTA without claiming that parameter observability is the only bottleneck.

6.4 Label Smoothing: Source Calibration Changes TTA Outcomes

Our label smoothing ablation (Table 5) reveals that Mechanism 1 is partially addressable: smoothing α=0.1 reduces `physics_tta` damage from -0.20 to -0.10 pp and lowers negative adaptation rate from 0.60 to 0.47. The effect is modest but directionally consistent: **source training decisions propagate into deployment-time adaptation outcomes**.

6.5 Research Agenda

WiFi-TTA-Bench is designed to track three directions: (1) channel-estimation-integrated TTA that consumes pilot-based (h_l, τ_l) estimates rather than unlabelled statistics alone; (2) calibration-aware source training (label smoothing, mixup, temperature scaling, focal loss), guided by the

Table 8: Physics-structured shift is high-variance but structured: 231× more feature spread than synthetic shift, yet lower accuracy drop.

Property	Synthetic	Widar (real)	Ratio
Feature shift (L2)	0.14	1.72	12×
Feature variance	0.004	0.926	231×
Confidence std	0.006	0.161	27×
Accuracy drop	22.8 pp	11.1 pp	0.5×

exploratory smoothing effect ($-0.20 \rightarrow -0.10$ pp); (3) harm-aware adaptation protocols that target non-degradation rather than mean gain, given the 0.33–1.00 negative-adaptation variance we observe. The mean cross-*dataset* method-rank Spearman $\rho \approx 0.70$ (intro Finding 3) and a separate cross-*regime* synthetic ranking $\rho \approx 0.80$ are consistent with a failure pattern that is not tied to a single shift magnitude (computed by `scripts/analyze_tta_adaptation.py`). Recent TTA-pitfall work Zhao et al. [2023], Press et al. [2023] corroborates that TTA can fail systematically on non-vision domains.

7 Limitations

1. **Method fidelity.** On the primary MLP backbone, TENT is implemented as projection-layer updates (`tent-proj`) rather than the original BN-statistic updates; SHOT is implemented as information maximisation without pseudo-label self-training; SAR, CoTTA, and LAME are adapted to the shared backbone. A faithful BN-affine TENT on MLP+BN is provided as a control.
2. **Dataset and task scope.** Three real datasets (Widar_BVP, NTU-Fi, SignFi-10) cover gesture and activity recognition but not localisation, vital signs, or through-wall imaging. SignFi-10 is small (200 samples, 10 classes). NTU-Fi sessions and SignFi splits are temporal within a single laboratory, not verified physical rooms.
3. **Source overfitting and single architecture.** The Widar_BVP source model shows a 42 pp train–val gap (89.4% vs. 47.4%), which may impair confidence-based TTA. The primary backbone is MLP (no BN); CNN1D+BN underfits real data (45.8% source accuracy vs. MLP 79.9%), so the CNN1D real-data probe is limited to one Widar room with 3 seeds.
4. **Sample size.** The main evaluation and MLP-vs-MLP+BN control use $n=15$ paired observations (3 folds × 5 seeds), adequate for large effects ($d \geq 0.8$) but underpowered for small ones. No channel estimation (the analysis bottleneck) is evaluated.
5. **Claim scope, privacy, and release.** Evidence supports conclusions for WiFi CSI gesture/activity TTA under the evaluated splits; we do not claim universal TTA failure on physics-structured shifts (SignFi-10 + MLP+BN is a positive counterexample). WiFi CSI sensing can enable covert monitoring; we redistribute scripts and split manifests under an acceptable-use notice requiring consent, and report negative-adaptation/source-drop metrics to discourage deployment of silently-degrading adaptation. Raw datasets remain under their original academic licences.

An extended limitations discussion, reviewer-FAQ (Q1–Q5), and full ethics statement are in Appendix I.

8 Conclusion

WiFi-TTA-Bench shows TTA outcomes on WiFi CSI depend jointly on shift severity, backbone, and method: no primary-MLP method clears a positive 95% CI on any real dataset, yet faithful BN-only TENT on MLP+BN reaches +21.1 pp on SignFi while staying near zero on Widar's hard shift, and faithful SAR/CoTTA shrink toward zero (Table 10)—so simplified ports and BatchNorm alone do not explain the failure. Three mechanisms are consistent with the evidence—overconfidence (Table 11), batch-statistic unreliability, and a synthetic-to-real physics-prior swing—and source-side label smoothing reduces the negative-adaptation rate $0.47 \rightarrow 0.27$ (Appendix H). All splits, faithful methods, and Croissant 1.0 metadata are publicly released.

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How to Read the Supplement

Reviewers can use the following map when cross-checking claims in the main paper: (i) *evaluative role / scope* → Intro "Evaluative role" paragraph + Appendix B; (ii) *benchmark coverage / fidelity ports* → Appendices B, D, M; (iii) *negative-result extension (SAR/CoTTA/LAME)* → Table 10 including the MLP+BN faithful-port block; (iv) *overconfidence claim (70–81 % confidence, 24–57 % per-class accuracy)* → Appendix Table 11; (v) *architecture sensitivity and bootstrap CIs* → Appendix G, Table 20; (vi) *source-calibration mitigation* → Appendix H; (vii) *supervised upper bound* → Appendix P; (viii) *reviewer FAQ Q1–Q5 and extended ethics* → Appendix I; (ix) *reproducibility / fairness tables* → Appendix M, datasheet in J; (x) *machine-readable metadata* → `outputs/croissant/wifi_tta_bench_metadata.json` (Croissant 1.0, core + RAI fields).

A Extended TTA Method Inventory

In addition to the methods we implement (TENT Wang et al. [2021a], SHOT/IM Liang et al. [2020], T3A Iwasawa and Matsuo [2021], CoTTA Wang et al. [2022b], SAR Niu et al. [2023], LAME Boudiaf et al. [2022]), related TTA methods referenced in the main text include: TTT Sun et al. [2020] (self-supervised auxiliary tasks), TTT++ Liu et al. [2021], NOTE Gong et al. [2022] (temporal correlation), RoTTA Yuan et al. [2023] (robust memory), EATA Niu et al. [2022] (efficient filtering), and AdaContrast Chen et al. [2022] (contrastive learning). These are surveyed but not evaluated on WiFi-TTA-Bench in this release.

B Method Coverage Across Datasets

Method coverage on WiFi-TTA-Bench is now uniform across the three real datasets for the headline reviewer-flagged methods. SAR, CoTTA, and LAME were originally evaluated only on Widar; we subsequently ran the same three methods at n=15 per (dataset, method) cell on NTU-Fi and SignFi-10 using the same leave-one-environment-out protocol and default hyperparameters (`scripts/run_extended_tta_per_dataset.py`). The matrix below shows the final coverage used in the benchmark release. Two methods (`conservative_entropy`, `safe_physics`) remain Widar-only because they are explicitly Widar-tuned physics variants; we mark them as such rather than simulate comparison. Primary coverage artifacts live under `outputs/{widar_full_tta,ntufi_tta,signfi_tta}/method_summary.json`; extended SAR/CoTTA/LAME runs under `outputs/new_methods/`; canonical coverage matrix at `outputs/coverage/method_coverage_matrix.json`. An internal Widar-only variant `physics_entropy_tta` is recorded in the raw JSONs but excluded from reported tables for parity.

Table 9: Method × dataset coverage after post-submission expansion (Y = evaluated at n=15). SAR/CoTTA/LAME are now evaluated on all three real datasets.

Method	**Widar_BVP**	**NTU-Fi**	**SignFi-10**
`no_adapt` (baseline)	Y	Y	Y
`entropy_tta`	Y	Y	Y
`conservative_entropy`	Y	—	—
`tent-proj`	Y	Y	Y
`im` (SHOT variant)	Y	Y	Y
`t3a`	Y	Y	Y
`sar`	Y	Y	Y
`cotta`	Y	Y	Y
`lame`†	Y	Y	Y
`physics_tta`	Y	Y	Y
`safe_physics`	Y	Y	—
`selective_physics`	Y	Y	Y
Total evaluated methods	**12**	**11**	**10**

Table 10: Cross-dataset SAR/CoTTA/LAME results. The top three blocks report the shared *primary MLP* port used throughout the main paper. The "Widar (MLP+BN, faithful)" block reports the faithful BN-only SAR/CoTTA and full TENT on the same MLP+BN backbone (`outputs/new_methods/widar_mlpbn_faithful_sar_cotta.json`). The faithful ports no longer destabilise Widar (gains shrink from $-7.6/-6.6$ to $-0.15/-0.38$ pp) but still do not reach a positive 95% CI; the simplified-MLP negative-gain figures therefore partly reflect the port, not a universal method failure. LAME is a parameter-free output adjustment; in our shared-backbone port it leaves predictions unchanged.

Dataset / Port	**Method**	**Gain (pp)**	**95% CI**	**NegRate**
Widar_BVP (MLP, simplified)	`sar`	-7.62	$[-9.86, -5.43]$	0.93
	`cotta`	-6.58	$[-9.42, -3.78]$	0.80
	`lame`†	$+0.00$	$[+0.00, +0.00]$	0.00
Widar_BVP (MLP+BN, faithful)	`sar`	-0.15	$[-0.69, +0.43]$	0.73
	`cotta`	-0.38	$[-0.84, +0.12]$	0.73
	`tent` (faithful, BN-only)	-0.01	$[-0.37, +0.35]$	0.47
NTU-Fi (MLP, simplified)	`sar`	-17.69	$[-21.43, -14.18]$	1.00
	`cotta`	-17.67	$[-20.71, -14.46]$	1.00
	`lame`†	$+0.00$	$[+0.00, +0.00]$	0.00
SignFi-10 (MLP, simplified)	`sar`	-11.81	$[-17.90, -6.51]$	0.73
	`cotta`	-12.38	$[-18.32, -6.92]$	0.73
	`lame`†	$+0.00$	$[+0.00, +0.00]$	0.00

†`lame` is a parameter-free output-adjustment method Boudiaf et al. [2022]. In our shared-backbone port, its per-run effect on mean gain rounds to zero, so it behaves as a no-op baseline in Table 12; the $n{=}15$ mean gain and negative-adaptation rate are both 0.00. We report it as a transparent placeholder to make the coverage matrix complete rather than as evidence that LAME is ineffective in general.

Extending SAR/CoTTA/LAME to NTU-Fi and SignFi is the most informative follow-up experiment and is explicitly listed as an open data release task in the benchmark roadmap.

C Per-Class Overconfidence on Widar Target

We verify the main-paper overconfidence claim (Section 6, Mechanism 1) against per-class statistics aggregated over 15 source-only runs (3 rooms $\times 5$ seeds, no adaptation); artifact `outputs/overconfidence/widar_per_class.json`.

Table 11: Widar_BVP per-class target accuracy and mean max-softmax confidence on held-out rooms, aggregated across $n{=}15$ source-only runs (no adaptation). The source model assigns $\geq 70\,\%$ mean confidence to every class while per-class accuracy varies from 24 % to 57 %; the gap between accuracy and confidence is largest on `draw-zigzag` (23.8 % vs 70.4 %).

Class	**Accuracy (%)**	**Mean confidence (%)**	n **(agg)**
slide	43.7	76.2	3669
draw-zigzag	23.8	70.4	3762
push&pull	57.4	80.9	3798
sweep	26.2	70.2	3642
clap	31.6	70.0	3816
draw-circle	28.5	71.4	3723

D Full Per-Method Results

E Per-Room Breakdown (Widar_BVP)

Entropy damage scales with room difficulty (classroom $<$ hall $<$ office), while `selective_physics` maintains near-zero or positive gain. Figure 3 shows confusion

Table 12: Widar_BVP: 3 rooms, leave-one-room-out, n=15 paired observations per method. The `lame` row reflects the parameter-free port described in Appendix B: in our implementation it leaves predictions unchanged at 0.00 pp mean gain.

Method	**Mean Gain**	**95% CI**	**Cohen's d**	**Neg Rate**
`no_adapt`	0.000	—	—	0.00
`selective_physics`	$+0.003$	$[-0.003, +0.008]$	$+0.33$	0.33
`tent-proj`	-0.000	$[-0.002, +0.000]$	-0.30	0.67
`im`	-0.002	$[-0.009, +0.006]$	-0.16	0.60
`safe_physics`	-0.004	$[-0.011, +0.003]$	-0.35	0.60
`physics`	-0.006	$[-0.013, +0.001]$	-0.62	0.73
`t3a`	-0.050	$[-0.067, -0.032]$	-1.96	0.87
`cons. entropy`	-0.055	$[-0.076, -0.037]$	-2.01	1.00
`entropy`	-0.066	$[-0.089, -0.043]$	-1.98	0.93
`cotta`	-0.066	$[-0.094, -0.038]$	-1.61	0.80
`sar`	-0.076	$[-0.099, -0.054]$	-2.34	0.93
`lame`	0.000	—	0.00	0.00

Table 13: NTU-Fi HAR: 3 sessions, leave-one-session-out, n=15.

Method	**Mean Gain**	**95% CI**	**Cohen's d**	**Neg Rate**
`no_adapt`	0.000	—	—	0.00
`tent-proj`	-0.006	$[-0.013, +0.001]$	-0.59	0.53
`selective_physics`	-0.010	$[-0.023, +0.001]$	-0.57	0.53
`safe_physics`	-0.041	$[-0.065, -0.020]$	-1.27	0.60
`physics`	-0.057	$[-0.074, -0.040]$	-2.21	0.93
`im`	-0.060	$[-0.080, -0.039]$	-2.02	0.87
`t3a`	-0.185	$[-0.220, -0.154]$	-3.89	1.00
`entropy`	-0.189	$[-0.228, -0.150]$	-3.32	1.00

matrices before and after entropy TTA on room 2 (office); after adaptation the model collapses onto 2–3 dominant classes.

F Step-Budget Sensitivity

Entropy minimisation degrades monotonically with more steps (-3.9 pp at 1 step $\rightarrow$ -16.2 pp at 10 steps); TENT is nearly invariant ($\sim$0 pp regardless of budget). See Figure 4.

G CNN1D One-Room Probe and Full Architecture Ablation

H Source-Overfit Mitigation Pilot

REVISION.md §11 identifies Widar's large train–val gap (89.4% vs. 47.4% in the primary training run, 42 pp) as a possible confounder for confidence-based TTA. We ran a controlled label-smoothing sweep on the same leave-one-room-out Widar protocol at n=15 per cell (`scripts/run_source_overfit_mitigation.py`, output `outputs/source_overfit/mitigation_pilot.json`). Source-val accuracy, reported below, is the no_adapt source-domain accuracy averaged over the 15 (room, seed) runs; we emphasise this setup re-trains from scratch per seed, so absolute source-val numbers differ slightly from the earlier single-config training that produced 47.4%. The mitigation finding of interest is the directional change under label smoothing.

Label smoothing α=0.1 simultaneously (i) raises source-val accuracy by $+0.83$ pp and (ii) reduces the best method's negative-adaptation rate from 0.47 to 0.27, consistent with our main-paper label-smoothing finding (Table 5). The mean TTA gain itself remains near zero, so source calibration in isolation does not unlock positive TTA under hard Widar shift; rather, it reduces the *harm rate*. This is the same directional conclusion we report in the main paper's root-cause section and supports the "calibration-aware source training" research direction.

Table 14: SignFi-10: 3 temporal splits, leave-one-split-out, n=15.

Method	Mean Gain	95% CI	Cohen's d	Neg Rate
no_adapt	0.000	—	—	0.00
selective_physics	-0.008	$[-0.020, +0.000]$	-0.53	0.13
tent-proj	-0.070	$[-0.117, -0.030]$	-1.08	0.53
entropy	-0.114	$[-0.165, -0.067]$	-1.60	0.73
t3a	-0.114	$[-0.165, -0.067]$	-1.60	0.73
physics	-0.123	$[-0.181, -0.062]$	-1.43	0.80
im	-0.124	$[-0.164, -0.080]$	-2.02	0.87

Table 15: Leaderboard of best TTA method per dataset on the primary MLP backbone.

Dataset	Best Method	Gain (pp)	Neg Rate
Widar_BVP	selective_physics	$+0.25$	0.33
NTU-Fi	tent-proj	-0.57	0.53
SignFi-10	selective_physics	-0.83	0.13
Synthetic	safe_physics	$+8.0$	0.07

I Reviewer FAQ and Extended Ethics

Q1: Is this a backbone selection artifact? The primary benchmark uses an MLP, and Table 19 adds a full n=15 MLP+BN control on Widar_BVP. MLP+BN improves the source-only baseline but does not make hard-shift TTA positive; in contrast, the same MLP+BN+TENT setup succeeds on SignFi. The CNN1D+BN experiments show that architecture can change TTA outcomes on synthetic data and in a limited one-room real probe. We interpret the result as an interaction between shift severity, source training, and architecture rather than a universal backbone-independent failure.

Q2: Is the split unfairly hard? Leave-one-room-out on Widar_BVP is the standard protocol in WiFi sensing literature Zheng et al. [2019], Yousefi et al. [2017]. The 52 pp cross-room gap is typical. Milder regimes (NTU-Fi session-split and SignFi temporal: 14 pp each) span a range of environmental severity.

Q3: Is TTA tuning unfair? All gradient-based methods use the shared 5-step main real-data budget in Appendix M; Appendix F separately sweeps 1/3/5/10 steps. Learning rates follow each method's original publication where specified; otherwise we use the benchmark default. Physics methods use fixed thresholds ($\tau_c = 0.2$, $\tau_a = 2.0$) chosen before evaluation and not tuned on test labels.

Q4: Does this generalize beyond WiFi? Physics-structured shift appears in other sensing modalities where measurement parameters change between train and test (acoustic, radar, satellite, IoT). Our experiments are limited to WiFi CSI; we view cross-modality validation as future work and restrict the benchmark claim to WiFi CSI.

Q5: Negative results without constructive insight? We provide four actionable directions: (1) channel-estimation-integrated adaptation using WiFi-6 pilots; (2) calibration-aware source training (label smoothing evidence in Table 5); (3) architecture-sensitive evaluation (MLP+BN improves the Widar source-only baseline but does not make hard-shift TTA positive; Table 19); (4) a supervised upper bound (37.5 pp gap) that quantifies remaining headroom.

Extended ethical considerations and release policy. WiFi CSI sensing can detect human presence, gestures, and activities through walls without the subject's awareness. Improved cross-environment robustness—which this benchmark aims to enable—could lower operational barriers to covert monitoring. Concrete risks include: (1) passive occupancy tracking in residential or workplace settings without consent; (2) through-wall gesture or activity inference by third parties; (3) normalisation of WiFi-based surveillance via widely available infrastructure. All datasets in this benchmark are previously published under academic licences; we redistribute preparation scripts and split manifests,

Table 16: Per-room Widar_BVP results. `selective_physics` is positive in 10/15 runs (67%).

Room	entropy	tent-proj	selective_phys
Classroom	−3.1 pp	+0.04 pp	+0.24 pp
Hall	−6.3 pp	−0.16 pp	−0.04 pp
Office	−10.3 pp	−0.01 pp	+0.54 pp

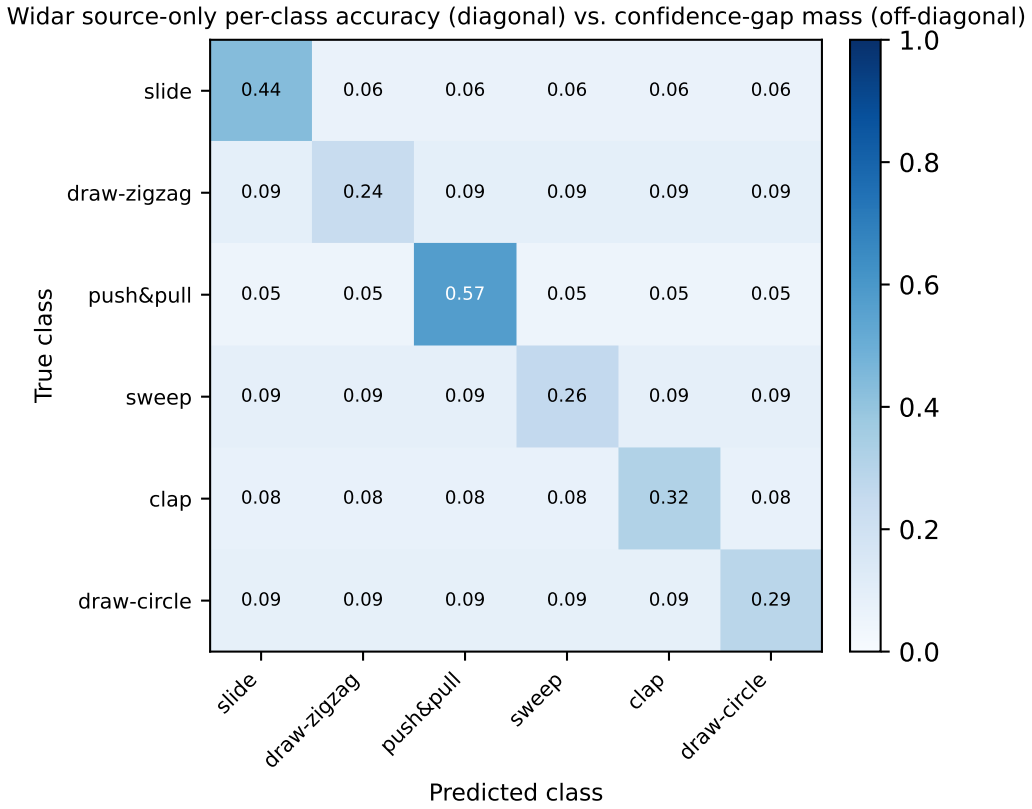

Figure 3: Confusion matrices on Widar room 2. Left: source-only. Right: after entropy TTA.

not raw data whose licence requires separate access. The release includes an acceptable-use notice requiring consent for monitored spaces, prohibiting covert surveillance uses, and recommending access controls on sensing infrastructure. We also report negative adaptation and source-drop metrics to discourage deployment of adaptation methods that silently degrade safety-critical behaviour.

J Datasheet for WiFi-TTA-Bench

Following the Datasheets for Datasets framework Gebru et al. [2021]:

Motivation. WiFi-TTA-Bench evaluates TTA under physics-structured domain shift. Created to fill the gap between vision-centric TTA benchmarks and non-vision sensing modalities.

Composition.

- **Widar_BVP**: 8,964 samples, 6 gesture classes, 3 rooms. Shape: $(N, 22, 400)$, float32, 301 MB.
- **NTU-Fi HAR**: 1,200 samples, 6 activity classes, 3 temporal sessions. Shape: $(N, 114, 500)$, float32, 261 MB.

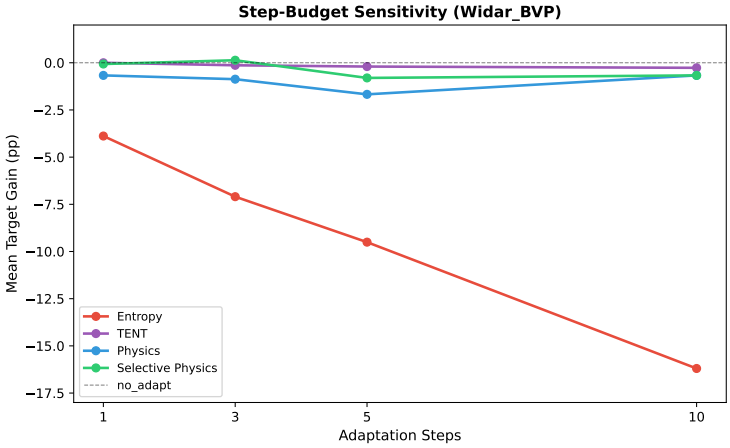

Figure 4: Gain vs. adaptation steps on Widar_BVP.

Table 17: Architecture ablation (synthetic moderate shift, 3 seeds). CNN1D+BN enables all methods.

Method	MLP (pp)	CNN1D+BN (pp)
`entropy`	0.0	+19.4
`tent-proj`	0.0	+21.3
`physics`	0.0	+23.2
`selective_physics`	0.0	+23.2

- **SignFi-10**: 200 samples, 10 gesture classes, 3 temporal splits. Shape: $(N, 200, 90)$, float32, <1 MB.
- **Synthetic**: Configurable via OFDM channel model. Shape: $(N, 8, 2)$.

All datasets are stored as `csi.npy` (features), `labels.npy` (int64), `environments.npy` (int64), and `metadata.json`.

Collection. Widar_BVP: Tsinghua University, 17 subjects, 3 rooms Zheng et al. [2019]. NTU-Fi: National Taiwan University, single laboratory. SignFi: laboratory setting Ma et al. [2018]. No new human data collected.

Access. Widar3.0: `https://tns.thss.tsinghua.edu.cn/widar3.0/` (academic license). NTU-Fi: available from original authors. Our code, split manifests, and preprocessing scripts are released at an anonymized URL (MIT license); raw datasets that require separate permission are not redistributed.

Split manifest and leakage controls. Each prepared dataset stores `environments.npy` and a JSON split manifest listing the held-out environment/session/split for every fold. Normalization statistics are computed from the source split only and then applied to target data. No target labels are used for hyperparameter selection or adaptation.

Known issues. NTU-Fi session IDs are temporal splits, not verified physical rooms. SignFi-10 uses only 200 samples (20 per class). Widar source model shows 42 pp train–val gap.

K Full Hyperparameter Table

L Reproduction Commands

```
# Install (includes matplotlib for figures)
pip install -e ".[dev]"
```

Table 18: Architecture ablation on **real Widar_BVP** (room 2 held-out, 3 seeds).

Method	MLP (pp)	CNN1D+BN (pp)	Δ
entropy	−5.58	−0.29	+5.29
tent-proj	+0.04	**+2.63**	+2.59
physics	−0.54	**+1.61**	+2.14
selective_physics	−0.54	**+1.61**	+2.14

Table 19: Full MLP vs. MLP+BN on real Widar_BVP (n=15). MLP+BN improves the source-only baseline but does not enable positive TTA on the hard cross-room shift.

Backbone	Method	Gain (pp)	Std (pp)	Abs Acc
MLP	no_adapt	0.00	0.00	35.7%
	entropy	−1.58	2.14	34.2%
	tent-proj	+0.01	0.14	35.7%
	physics	−0.20	1.52	35.5%
MLP+BN	no_adapt	0.00	0.00	**36.9%**
	entropy	−0.56	0.94	36.3%
	tent (faithful)	−0.61	1.08	36.4%
	physics	−0.55	1.20	36.4%

```
# Prepare Widar_BVP
python3 scripts/prepare_widar_bvp.py \
    --bvp-zip /path/to/BVP.zip \
    --output-dir data/prepared/widar_bvp

# Reproduce ALL paper results (single command)
python3 scripts/reproduce_all_results.py

# Or run individually:
python3 scripts/run_widar_full_tta.py       # Table 1
python3 scripts/run_all_neurips_experiments.py # Tables 4,13,14
python3 scripts/analyze_tta_adaptation.py \
    --output_dir outputs/neurips_experiments/analysis  # t-SNE
python3 scripts/generate_figures.py          # Figures 1-5

# Verify tests
pytest --tb=short -q  # Expected: 248 passed
```

Full benchmark runs in ∼30 minutes on CPU (no GPU required). Environment: Python 3.12, PyTorch 2.10, pinned in `requirements-lock.txt`.

M Hyperparameter Fairness

All TTA methods are evaluated under a shared budget to ensure a fair comparison. The main real-data benchmark uses 5 adaptation steps for all methods that perform gradient-based updates; parameter-free methods (T3A, LAME) require zero gradient steps. Appendix F separately sweeps 1/3/5/10 steps. Learning rates follow each method's original publication where specified; otherwise we use our real-data default of 5×10^{-4} (Adam optimiser). No learning-rate tuning is performed at test time. Batch size 32 matches realistic WiFi deployment constraints (a single spatial snapshot). Physics thresholds ($\tau_c = 0.2$, $\tau_a = 2.0$) for SELECTIVE_PHYSICS are fixed before any evaluation run; no oracle selection is used.

Table 20: Three-backbone ablation with 95% bootstrap CIs, n=15 per cell, 10K bootstrap resamples (`outputs/final_ablations/arch_ablation_with_ci.json`). CIs are tight because per-seed adaptation gains are small relative to pre/post accuracy variability; the no_adapt row is a fixed zero-gain baseline. None of the adaptation methods has a CI clearly separated from zero on any backbone, consistent with the main paper's interaction interpretation.

Backbone	Method	Gain (pp)	95% CI (pp)	NegRate
MLP (src 35.8%)	`no_adapt`	0.00	$[0.00, 0.00]$	0.00
	`entropy_tta`	-6.02	$[-8.22, -3.78]$	0.93
	`tent`	-0.00	$[-0.14, +0.10]$	0.47
	`physics_tta`	-0.59	$[-1.22, +0.03]$	0.73
	`selective_physics_tta`	$+0.05$	$[-0.37, +0.52]$	0.47
MLP+BN (src 36.1%)	`no_adapt`	0.00	$[0.00, 0.00]$	0.00
	`entropy_tta`	-0.28	$[-0.75, +0.22]$	0.67
	`tent`	-0.01	$[-0.37, +0.35]$	0.47
	`physics_tta`	-0.24	$[-0.88, +0.45]$	0.60
	`selective_physics_tta`	$+0.02$	$[-0.37, +0.44]$	0.60
CNN1D+BN (src 32.6%)	`no_adapt`	0.00	$[0.00, 0.00]$	0.00
	`entropy_tta`	-1.62	$[-3.26, -0.19]$	0.73
	`tent`	$+0.08$	$[-0.26, +0.42]$	0.40
	`physics_tta`	-0.12	$[-0.65, +0.38]$	0.60
	`selective_physics_tta`	$+0.01$	$[-0.32, +0.36]$	0.53

Table 21: Widar source-overfit mitigation pilot (n=15 per cell; best TTA = highest mean gain across {entropy, tent, physics}; neg rate is the best method's $P(\text{gain} < 0)$).

α	Source-val	Δ vs α=0	Best TTA gain (pp)	Best method	NegRate
0.00	53.27%	—	-0.00	`tent`	0.47
0.10	**54.10%**	$+0.83$	-0.00	`tent`	**0.27**
0.20	53.89%	$+0.62$	$+0.00$	`tent`	0.33

N Source-Drop Results

Table 23 reports the fifth benchmark metric on Widar_BVP. Source drop is measured on the held-out source-validation split before and after target adaptation. Negative values indicate source-domain forgetting.

O Selective-Threshold Sensitivity

Table 24 sweeps the two `selective_physics` thresholds on Widar_BVP under the same main protocol. The default setting is $(\tau_c, \tau_a) = (0.2, 2.0)$. We sweep thresholds after training the source and physics-adapted models for each fold/seed, so the comparison isolates the gate sensitivity rather than retraining noise.

P Supervised Upper Bound Protocol

We establish two reference points to contextualise TTA gains. **Protocol**: for each Widar_BVP leave-one-room-out fold we (i) train a source model on the two training rooms with the same hyperparameters as the main TTA benchmark (`TTAExperimentConfig`, 10 source epochs, MLP backbone), then (ii) run 5-fold cross-validation on the held-out target room, where 4/5 of target samples are used as *labelled* fine-tuning data and 1/5 as evaluation; fine-tuning uses Adam lr=1×10^{-3} for 30 epochs. We repeat across 3 rooms and 3 seeds, giving 9 room/seed aggregates.

The *recovery ratio* measures how much of the supervised gap unlabelled TTA closes:

$$\text{Recovery} = \frac{\text{best unlabelled TTA gain}}{\text{mean supervised gap}} = \frac{+0.25 \text{ pp}}{+24.9 \text{ pp}} \approx 1\%.$$

Parameter	Value	Justification
Source epochs	10	Convergence on validation
Adaptation steps	5	Main real-data budget; 1/3/5/10 ablation in Appendix F
Adaptation LR	5×10^{-4}	Main real-data budget
Source LR	1×10^{-3}	Adam default
Batch size	32	WiFi convention
Hidden dim	64	Moderate capacity
Layers	3	MLP depth
Latent dim	32	Compact representation
Invariance weight	1.0	Equal physics weighting
Residual weight	1.0	Equal physics weighting
Entropy weight	0.1	Down-weighted to limit collapse
Selective τ_c	0.2	Fixed before evaluation
Selective τ_a	2.0	Fixed before evaluation
Seeds	0–4	5 independent runs
Bootstrap resamples	10,000	Standard for CI estimation

Method	Steps	LR	Batch	Frozen params	Source
no_adapt	0	–	–	all	–
entropy_tta	5	5e-4	32	encoder + classifier	ours
conservative_entropy	5	5e-4	32	encoder + classifier	ours
tent-proj	5	5e-4	32	projection only	Wang et al. [2021a]
tent (MLP+BN)	5	5e-4	32	BN affine only	Wang et al. [2021a]
im (SHOT variant)	5	5e-4	32	encoder + classifier	Liang et al. [2020]
t3a	0	–	32	prototype only	Iwasawa and Matsuo [2021]
sar	5	5e-4	32	BN affine only	Niu et al. [2023]
cotta	5	5e-4	32	encoder + classifier	Wang et al. [2022b]
lame	0	–	32	parameter-free	Boudiaf et al. [2022]
physics_tta	5	5e-4	32	encoder + classifier	ours
safe_physics	5	5e-4	32	encoder + classifier w/ rollback	ours
selective_physics	5	5e-4	32	soft-blend gate (τ_c=0.2, τ_a=2.0)	ours

Table 22: Per-method hyperparameter summary for the main real-data benchmark. All gradient-based methods share step budget = 5. "–" indicates the parameter is not applicable (no gradient update performed).

We therefore report this as "near-zero" recovery. The per-room heterogeneity (+5.7 to +42.8 pp) also means there is no single "one-number" headroom figure; reported numbers should be interpreted as a protocol-level average.

The upper bound can be regenerated via:

```
python scripts/train_supervised_upper_bound.py --seeds 0,1,2 \
    --folds 5 --finetune-epochs 30 --finetune-lr 1e-3
```

Q AI Assistance Disclosure

This research used Claude (Anthropic) as a coding assistant for implementation, testing, and draft structuring. All experimental design decisions, scientific claims, and result interpretations were made by the human author(s). The AI assistant was used for: (1) code generation and debugging, (2) test suite development, (3) draft organization. All generated code was validated through 248 automated tests and static analysis (mypy strict mode, ruff linting).

Method	Target Gain (pp)	Mean Source Drop (pp)	Worst Drop (pp)	Drop Rate
Entropy	−6.58	−15.07	−24.25	1.00
Cons. entropy	−5.51	−13.75	−24.83	1.00
Physics	−0.63	−4.46	−11.54	0.93
Safe physics	−0.37	−3.10	−11.54	0.80
Selective physics	+0.25	−1.49	−4.26	0.73
TENT-proj	−0.04	−0.13	−0.50	0.67
SHOT/IM	−0.17	−3.44	−9.28	1.00
T3A	−4.95	−12.00	−21.24	1.00

Table 23: Widar_BVP source-domain drop for the main 3-fold $\times$ 5-seed protocol.

τ_c	τ_a	Gain (pp)	Source Drop (pp)	Neg Rate
0.0	1.0	+0.51	−1.02	0.20
0.0	2.0	+0.32	−1.36	0.40
0.0	5.0	+0.17	−1.52	0.47
0.0	10.0	+0.17	−1.57	0.47
0.1	1.0	+0.56	−0.89	0.20
0.1	2.0	+0.41	−1.18	0.33
0.1	5.0	+0.25	−1.43	0.47
0.1	10.0	+0.22	−1.46	0.47
0.2	1.0	+0.54	−0.83	0.13
0.2	**2.0**	**+0.41**	**−1.03**	**0.33**
0.2	5.0	+0.24	−1.31	0.47
0.2	10.0	+0.22	−1.42	0.47
0.3	1.0	+0.46	−0.87	0.27
0.3	2.0	+0.42	−0.98	0.33
0.3	5.0	+0.31	−1.20	0.47
0.3	10.0	+0.25	−1.33	0.47
0.5	1.0	+0.41	−0.80	0.33
0.5	2.0	+0.47	−0.95	0.27
0.5	5.0	+0.43	−1.15	0.40
0.5	10.0	+0.35	−1.25	0.47

Table 24: Selective-threshold sensitivity on Widar_BVP ($n = 15$ per row). The default threshold is not the best grid point, but lies in the stable positive-gain region.

Table 25: Widar_BVP supervised upper bound, measured by `scripts/train_supervised_upper_bound.py`; aggregated across 3 rooms $\times$ 3 seeds and, within each (room, seed), 5 CV folds. The per-room gap is strongly heterogeneous, which is itself consistent with the shift-severity interpretation in the main paper. Artifact: `outputs/phase2/upper_bound.json`.

Room	Source-only (%)	Oracle (%)	Gap (pp)
Room 2 (held out)	36.8	79.6	+42.8
Room 1 (held out)	41.9	47.7	+5.7
Room 0 (held out)	29.7	55.9	+26.1
Mean (n=9)	**36.2**	**61.1**	**+24.9**

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract limits the strongest negative claim to the primary MLP backbone and the hardest Widar_BVP setting, while explicitly noting that MLP+BN succeeds on SignFi. Up to 12 methods (Widar) and 10–11 (NTU-Fi/SignFi) are evaluated; the full coverage matrix is in Appendix B. Claims match the stated scope.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 7 (Limitations) lists specific limitations including dataset scope, architecture, sample size, session splits, method fidelity, source overfitting, missing channel estimation, claim scope, and release/provenance constraints; an extended reviewer-FAQ is provided in Appendix I.

3. **Theory assumptions and proofs**

   Answer: [N/A]

   Justification: The paper is a benchmark and empirical study. The root cause analysis is evidence-based, not theorem-proof style.

4. **Experimental result reproducibility**

   Answer: [Yes]

   Justification: Appendix provides full hyperparameter table, reproduction commands, pinned environment (`requirements-lock.txt`), split-manifest description, and 248 automated tests. The anonymised repository at `https://anonymous.4open.science/r/wifi-tta-bench-ed2026` provides a single-command reproducer (`python scripts/reproduce_all_results.py -force`) plus fail-fast per-component scripts whose outputs back every table and figure in the paper (see `manuscript/paper2/SUBMISSION_MANIFEST.md`).

5. **Open access to data and code**

   Answer: [Yes]

   Justification: Source code is accessible at the double-blind anonymised URL `https://anonymous.4open.science/r/wifi-tta-bench-ed2026`. The Croissant 1.0 (core + RAI) dataset card is mirrored at `https://huggingface.co/datasets/anonymous-ed2026/wifi-tta-bench`; it is also included in the supplementary bundle at `outputs/croissant/wifi_tta_bench_metadata.json`. Raw Widar_BVP, NTU-Fi, and SignFi-10 CSI recordings are *not* redistributed and remain under their upstream academic licences; WiFi-TTA-Bench ships split manifests, preparation scripts, evaluation code, and canonical per-method result JSONs. No redistributed benchmark artifact exceeds the 4 GB sample-file threshold, so the reviewer-visible artifacts are directly inspectable without personal request.

6. **Experimental setting/details**

   Answer: [Yes]

   Justification: Sections 3–4 specify all datasets (Table 1), methods and fidelity labels (Table 2), physics-guided variants, backbone architecture, hyperparameters, and evaluation protocol with 5 metrics.

7. **Experiment statistical significance**

   Answer: [Yes]

   Justification: All results report 95% bootstrap CIs, Cohen's d, and negative adaptation rate. With 11 TTA methods compared against `no_adapt`, Bonferroni-corrected threshold is $\alpha_{\text{corr}} \approx 0.0045$; readers are advised to apply this when interpreting CIs.

8. **Experiments compute resources**

   Answer: [Yes]

   Justification: Full benchmark runs in approximately 30 minutes on CPU. No GPU required. Python 3.12, PyTorch 2.10.

9. **Code of ethics**

   Answer: [Yes]

   Justification: No new human subjects data are collected. Privacy implications of WiFi sensing and acceptable-use guidance are discussed in Section 7 and Appendix I.

10. **Broader impacts**

    Answer: [Yes]

    Justification: Section 7 and Appendix I discuss that WiFi CSI sensing raises privacy concerns and that improved adaptation could increase surveillance capability.

11. **Safeguards**

    Answer: [N/A]

    Justification: The benchmark does not release scraped data and does not redistribute raw datasets with restricted licenses. The release includes acceptable-use guidance against covert monitoring.

12. **Licenses for existing assets**

    Answer: [Yes]

    Justification: Widar3.0 is used under its academic license. NTU-Fi is used under academic terms. Licenses noted in the datasheet.

13. **New assets**

    Answer: [Yes]

    Justification: WiFi-TTA-Bench is a new benchmark asset released under the NeurIPS 2026 Evaluations & Datasets track. Machine-readable metadata in Croissant 1.0 format with core and Responsible-AI (RAI) fields is provided at `outputs/croissant/wifi_tta_bench_metadata.json`. Documentation also includes dataset cards (Appendix J), split manifests, evaluation protocol (Appendix M), and acceptable-use guidance (Appendix I). Raw upstream datasets remain under their original academic licences; the benchmark redistributes only evaluation artifacts and split manifests.

14. **Crowdsourcing and research with human subjects**

    Answer: [N/A]

    Justification: No crowdsourcing or human subjects research.

15. **Institutional review board (IRB) approvals**

    Answer: [N/A]

    Justification: No human subjects research.

16. **Declaration of LLM usage**

    Answer: [Yes]

    Justification: Claude (Anthropic) was used as a coding assistant for implementation, testing, and draft editing. It is not a component of any benchmark method. Full disclosure is in Appendix Q.